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An evaluation of the severe acute respiratory infection surveillance system in Ireland

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Abstract

Background Expansion of the current single hospital site Severe Acute Respiratory Infection (SARI) surveillance programme in Ireland, to create a SARI sentinel hospital network, commenced in 2023. This evaluation aimed to assess the performance of key surveillance system attributes to improve and guide expansion.

Methods A mixed quantitative and qualitative evaluation was conducted from September 2021 to November 2022 using European Centre for Disease Prevention and Control (ECDC) guidance as a framework. SARI surveillance data were analysed to assess completeness and timeliness. Semi-structured focus group discussions and key informant interviews and an online stakeholder questionnaire, evaluated usefulness, simplicity, flexibility and acceptability. Qualitative data were analysed using thematic analysis.

Results Completeness of key variables including age and outcome was 100% ($n=747$). COVID-19 vaccination status was completed in 89%, completeness of ethnicity data was only 0.6%. Median time from hospitalisation to commencement of data entry was one day (IQR 1–3). Time to receipt of whole genome sequencing (WGS) results was 40 days (IQR 30–61). Themes identified in the qualitative evaluation were the complexity of SARI surveillance, dependence on key individuals and skilled staff. Suggested areas for improvement included process automation and additional resources to improve timeliness and acceptability, particularly during epidemic or pandemic surge periods.

Conclusions The SARI surveillance system reports complete and timely data. However, improvement in the timeliness of WGS data and recording of vaccination status and equity stratifiers e.g. ethnicity, would strengthen data quality. Priorities for expansion include process automation and adequate resource allocation, to enable reliable SARI surveillance at all times.

Keywords Public health monitoring, Pandemic preparedness, Information for action, Sentinel surveillance network

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Table 1 Methods used to evaluate surveillance attributes

Attribute 1: Sensitivity			
Indicator	**Type of data**	**Data collection method(s)**	**Data source(s)**
Case definition review	Quantitative	Analysis of SARI surveillance database comparing sensitivity of the WHO case definition to the current case definition* and a web-based questionnaire assessing opinion on case definition sensitivity	SARI surveillance database Additional stakeholders
	Qualitative	Focus group discussions and key informant interviews: Opinion on case definition sensitivity	Key stakeholders
Attribute 2: Data quality/internal completeness			
Indicator	**Type of data**	**Data collection method(s)**	**Data source(s)**
Number and percent with completed key variables for all SARI cases	Quantitative	Analysis of SARI surveillance database	SARI surveillance database
Number and percent with completed key variables for all SARI cases who have been discharged or who have died	Quantitative	Analysis of SARI surveillance database	SARI surveillance database
Opinion on data quality	Qualitative	Focus group discussion and key informant interviews	Key stakeholders
Attribute 3: Timeliness			
Indicator	**Type of data**	**Data collection method(s)**	**Data source(s)**
Median number of days (IQR) between key steps in the surveillance pathway	Quantitative	Analysis of SARI surveillance database	SARI surveillance database
Timeliness of producing reports for HPSC and reporting data to ECDC/The European Surveillance System (TESSy)	Quantitative	Document review	Weekly reports for HPSC Data reporting to TESSy
Whether reporting is timely and frequent enough to be useful to those who receive the report	Quantitative	Web-based questionnaire	Additional stakeholders
Attribute 4: Simplicity			
Indicator	**Type of data**	**Data collection method(s)**	**Data source(s)**
The simplicity of surveillance system processes	Qualitative	Focus group discussion and key informant interviews	Key stakeholders
Existence of a structured/standardised approach to data collection	Qualitative	Document review and focus group discussion and key informant interviews	Surveillance protocol, data flow diagram and key stakeholders
System flexibility (also evaluated as part of attribute 5)	Qualitative	Focus group discussion and key informant interviews	Key stakeholders
Attribute 5: Flexibility			
Indicator	**Type of data**	**Data collection method(s)**	**Data source(s)**
The ability of the system to easily adapt to new challenges and changes	Qualitative	Focus group discussion and key informant interviews	Key stakeholders
Assessment of completeness and timeliness as indicators of flexibility	Quantitative	Analysis of SARI surveillance database	SARI surveillance database
Attribute 6: Acceptability			
Indicator	**Type of data**	**Data collection method(s)**	**Data source(s)**
Perception of the surveillance system as important	Qualitative	Focus group discussion and key informant interviews	Key stakeholders
	Quantitative	Web based questionnaire	Additional stakeholders
Identification of obstacles to participation	Qualitative	Focus group discussion and key informant interviews	Key stakeholders
Assessment of completeness and timeliness as indicators of acceptability	Quantitative	Analysis of SARI surveillance database	SARI surveillance database
Attribute 7: Usefulness			
Indicator	**Type of data**	**Data collection method(s)**	**Data source(s)**
Usefulness of outputs	Quantitative	Web based questionnaire	Additional stakeholders
	Qualitative	Focus group discussions and key informant interviews	Key stakeholders

Table 1 (continued)

Assessment of completeness and timeliness as indicators of usefulness	Quantitative	Analysis of SARI surveillance database	SARI surveillance database
Assessment of whether the surveillance system is meeting its objectives	Qualitative	Focus group discussions and key informant interviews	Key stakeholders
	Quantitative	Web based questionnaire	Additional stakeholders

*At the time of the evaluation, an adapted version of the ECDC clinical case definition for a hospitalised possible COVID-19 case was used for SARI surveillance in Ireland [1, 2].

Introduction

Severe Acute Respiratory Infection (SARI) surveillance systems monitor the circulation, severity, and impact of respiratory viruses. Data collected can provide important early warning information to inform public health and health service preparedness and planning in the context of respiratory disease epidemics and pandemics [1–4]. SARI data are also used to measure vaccine and antiviral effectiveness [4–6].

In 2020, the European Centre for Disease Prevention and Control (ECDC) accelerated the development of a European-level SARI surveillance system. In Ireland, in July 2021, an active, case-based surveillance system, coordinated by the Health Protection Surveillance Centre (HPSC), was established in one tertiary care adult hospital [4].

Expansion of SARI surveillance in Ireland to include additional hospital sites and to create a SARI sentinel hospital surveillance network commenced in 2023. Prior to expansion, an evaluation of the existing SARI surveillance system was undertaken. The aim of the evaluation was to assess the performance of key SARI surveillance system attributes to identify areas for improvement within the existing system to inform and guide SARI surveillance expansion.

Methods

Study design

A mixed quantitative and qualitative evaluation was designed using guidance from the ECDC, the World Health Organization (WHO) and the Centers for Disease Control and Prevention (CDC) [7–9]. The following attributes were evaluated: data quality/internal completeness, timeliness, sensitivity of the case definition, simplicity, flexibility, acceptability and usefulness.

There were three components to the evaluation: analysis of data within the SARI surveillance system database, a qualitative evaluation and a web-based questionnaire. There was an overlap between the methodologies used to assess attributes (Table 1).

Study population

Two broad groups of stakeholders were identified; key stakeholders and additional stakeholders. Key stakeholders were defined as those who contribute to the reporting, collection, analysis, interpretation, and dissemination of surveillance data. Additional stakeholders were defined as those who receive the weekly national SARI surveillance report and who may use this information for the clinical or public health management of respiratory pathogens and/or to inform health service planning within the Irish health service and public health interventions.

Data collection and analysis

Evaluation component 1

Data were extracted from the SARI surveillance system on 9th November 2022 and were used to assess the attributes of data quality/internal completeness, timeliness, and sensitivity of the case definition between 13th September 2021 and 6th November 2022 (Table 1). All data were anonymised. Key variables and key steps in the surveillance pathways were identified from a review of the SARI surveillance protocol and the weekly SARI surveillance report [10], and following consultation with key stakeholders who were directly involved in SARI data collection at the hospital site and with HPSC SARI surveillance leads.

Completeness was assessed by examining the proportion of completed data fields for key variables collected on admission for all SARI cases and key variables for all SARI cases that had completed their episode of care, i.e., had been discharged or died. Fields left blank (where data were eligible to be entered), "do not know" and "not specified" responses were considered not completed.

The median number of days and inter quartile range (IQR) between key steps in the surveillance pathway were calculated to assess timeliness. Overall timeliness for the evaluation period and the timeliness for each month was calculated.

At the commencement of SARI surveillance in Ireland, the WHO SARI case definition was used [4]. However, due to concerns that this case definition was not sensitive enough, the case definition was changed [4]. At the time of this evaluation, an adapted version of the ECDC clinical case definition for a hospitalised possible COVID-19 case was used [4, 11]. This adapted ECDC case definition was a person hospitalised for at least 24 h with acute respiratory infection, with at least one of the following symptoms: cough, fever, shortness of breath OR sudden onset of anosmia, ageusia or dysgeusia, with onset of symptoms within 14 days prior to hospital admission.

To estimate the sensitivity of the WHO case definition compared to the adapted ECDC case definition [12, 13], all SARI cases were reclassified to the WHO case definition. The proportion of reported SARI cases that met the WHO case definition was calculated [12, 13].

Evaluation component 2

A qualitative evaluation using a combination of five semi-structured key informant interviews and two focus group discussions in December 2022 and January 2023, assessed the attributes of simplicity, flexibility, and acceptability. The qualitative component also evaluated data quality/internal completeness, usefulness, and the sensitivity of the case definition. Purposive sampling was undertaken among key SARI stakeholders, all key stakeholders from the participating hospital site, the HPSC and the National Virus Reference Laboratory were intentionally selected and invited to participate in the study based on their knowledge, experience and involvement in the SARI surveillance system. An interview guide was developed (Supplementary materials). The decision to conduct a focus group or key informant interview was based on the number of stakeholders available to participate from each distinct stakeholder group. If three or more participants were available, a focus group discussion was undertaken, otherwise a key informant interview was conducted [14]. Non-response and non-participation were recorded.

The lead evaluator (LM) was a female medical doctor with training in Public Health Medicine and General Practice who at the time of the evaluation was a fellow of the ECDC Fellowship Programme working in the HPSC who had not been working with the SARI surveillance programme prior to the evaluation.

The lead evaluator conducted all focus group discussions and key informant interviews for approximately one hour either face to face or online via video conferencing. Focus group discussions were recorded with consent and transcribed verbatim. Detailed notes were taken with consent for key informant interviews. Participants were informed that no comments would be directly attributed to any named individual participant; pseudonyms (e.g. FGD 2, P3) were assigned to ensure confidentiality.

Thematic analysis was the analytical approach used [15, 16]. Interview notes and transcripts were reviewed multiple times by the lead evaluator prior to transcribing to ensure familiarisation with the data. The theoretical framework for the data analysis was developed based on the surveillance system attributes under investigation using ECDC guidelines for surveillance systems evaluation [9]. Themes were identified deductively based on these guidelines. Within each theme, sub-themes were identified inductively and coded with direct quotes used to support the sub-themes identified (Supplementary materials). Interview transcripts and detailed written interview notes were single coded and sub-themes were identified by the lead evaluator. The consolidated criteria for reporting qualitative research (COREQ) guidelines were used for reporting [17].

Evaluation component 3

A web-based questionnaire was designed primarily to evaluate usefulness (which included whether the system was meeting its objectives). The sensitivity of the case definition, acceptability and timeliness were also evaluated. The questionnaire included questions with Likert scales, yes/no questions, and free text responses. Participants were informed of the purpose of the evaluation and all data were anonymised (Supplementary materials).

Prior to circulation, a pilot study was conducted among Epidemiologists and Consultants/Specialists in

Public Health Medicine in the HPSC ($n=7$). Following this, additional free-text response questions in relation to how recipients use the weekly report, surveillance objectives and sensitivity of the case definition were added.

A link to the questionnaire was circulated via email to those classified as additional stakeholders. Absolute frequencies and percentages for Likert scale and yes/no responses were calculated using non missing values as denominators. All free-text responses were categorised based on the surveillance attribute that was being assessed by the question.

All quantitative analysis was conducted using R version 4.2.1 [18].

Results

Participant characteristics

The professional backgrounds of participants (key stakeholders) in the two focus group discussions ($n=3$ participants per group) and the five key informant interviews varied and consisted of Epidemiologists, Surveillance Officers, Research Nurses, Medical Consultants in Emergency Medicine, Clinical Microbiology, Infectious Diseases, Consultants/Specialists in Public Health Medicine and Clinical Laboratory Scientists. Their roles in SARI surveillance also varied; some were involved in data collection, others in specimen retrieval and testing and/or data collation and reporting. Other stakeholders had an advisory, strategic or oversight role and were involved in establishing the SARI surveillance system.

There were 75 valid responses to the web-based questionnaire circulated to additional stakeholders ($n=362$), giving a response rate of 21% (75/362) (Table 2).

Data quality/internal completeness

Among all SARI cases ($n=747$) and SARI cases with a completed episode of care ($n=627$), completeness of data entry for the majority of key variables exceeded 90%. Completeness was lower for both self-reported and verified vaccination status variables and for drug prophylaxis. Completeness of the ethnicity variable was 0.6% (Fig. 1).

Timeliness

Commencement of clinical data entry, laboratory specimen collection and notification were timely (Fig. 2). The median time from hospital admission to completion of clinical data entry was 48 days (IQR 21–81), while the median length of hospital stay for SARI patients was 6 days (IQR 3–13). There was a median interval of 40 days (IQR 31–60) from hospital admission to receipt of whole genome sequencing (WGS) results for SARS-CoV-2 samples and 62 days (IQR 46–97) for influenza samples (Fig. 2).

In December 2021 and January 2022, during a surge period of SARI, median time intervals for commencement of data entry, notification, reference laboratory sample dispatch for sequencing and receipt of sequencing results increased (Supplementary materials).

Simplicity

Sub-theme: complex processes and operational challenges

All participants in the qualitative evaluation reported that the SARI surveillance system is a complex system that requires multiple manual processes to facilitate case identification, sample retrieval, data collection, collation,

Table 2 Characteristics of respondents to web-based questionnaire by professional group

	Characteristics ($n=75$)	*N*	%
Work location	Acute hospital	31	41.3
	Department of Public Health	23	30.7
	HPSC	13	17.3
	Department of Health	2	2.7
	Other work location*	4	5.3
	Not specified	2	2.7
Professional group	Epidemiologist/Surveillance scientist	17	22.7
	Consultant Microbiologist/Virologist	16	21.3
	Consultant/Specialist in Public Health Medicine**	12	16.0
	Consultant in Anaesthesiology/ICU Medicine	7	9.3
	Consultant in Emergency Medicine	6	8.0
	Senior Medical Officer	5	6.6
	Specialist Registrar in Public Health Medicine	1	1.3
	Nurse/Nurse Specialist or Manager	3	4.0
	Officer in the Department of Health	2	2.7
	Consultant Paediatrician	2	2.7
	Other professional group***	2	2.7
	Not specified	2	2.7

*Other work location includes academic posts and the COVID-19 contact management programme ** Includes Area Directors of Public Health ***Other professional group includes academic and lecturing posts

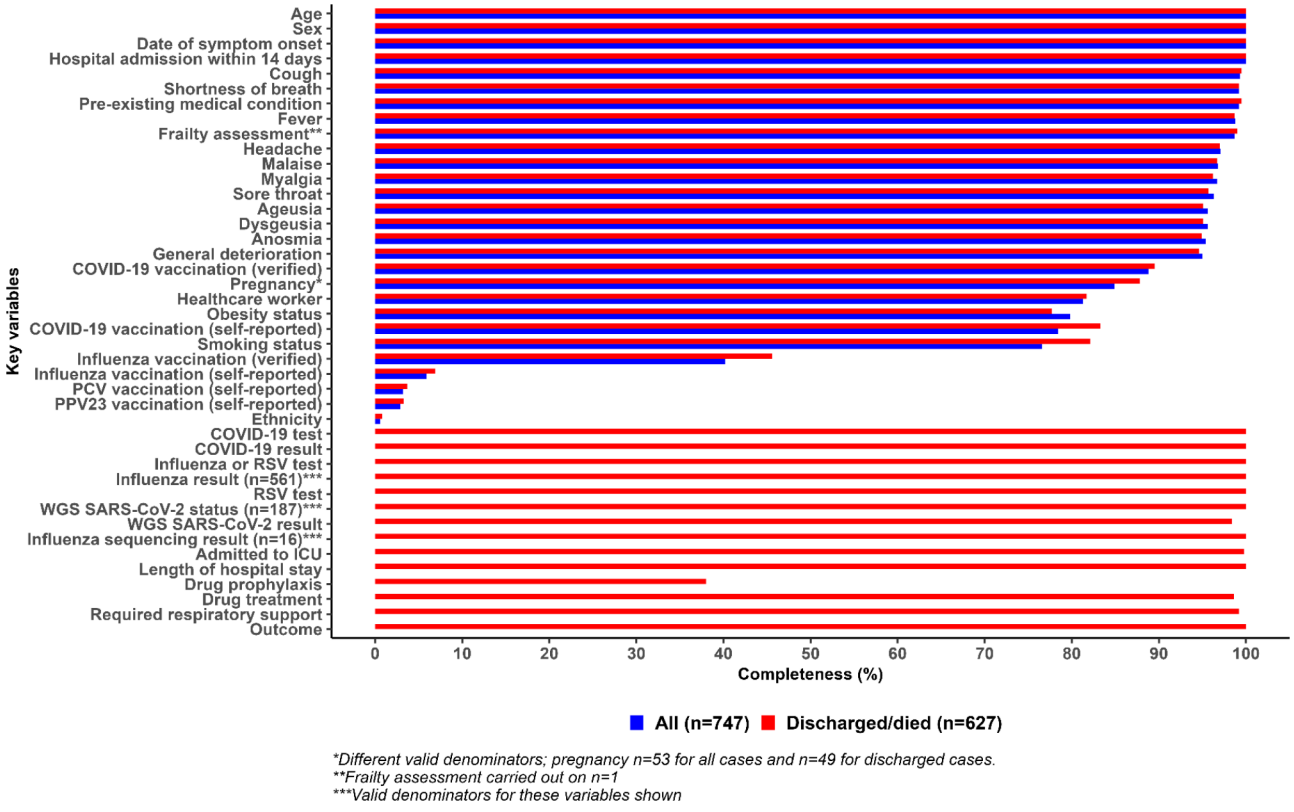


Fig. 1 Percentage completeness of key SARI variables: Among all SARI cases and those who had been discharged or died

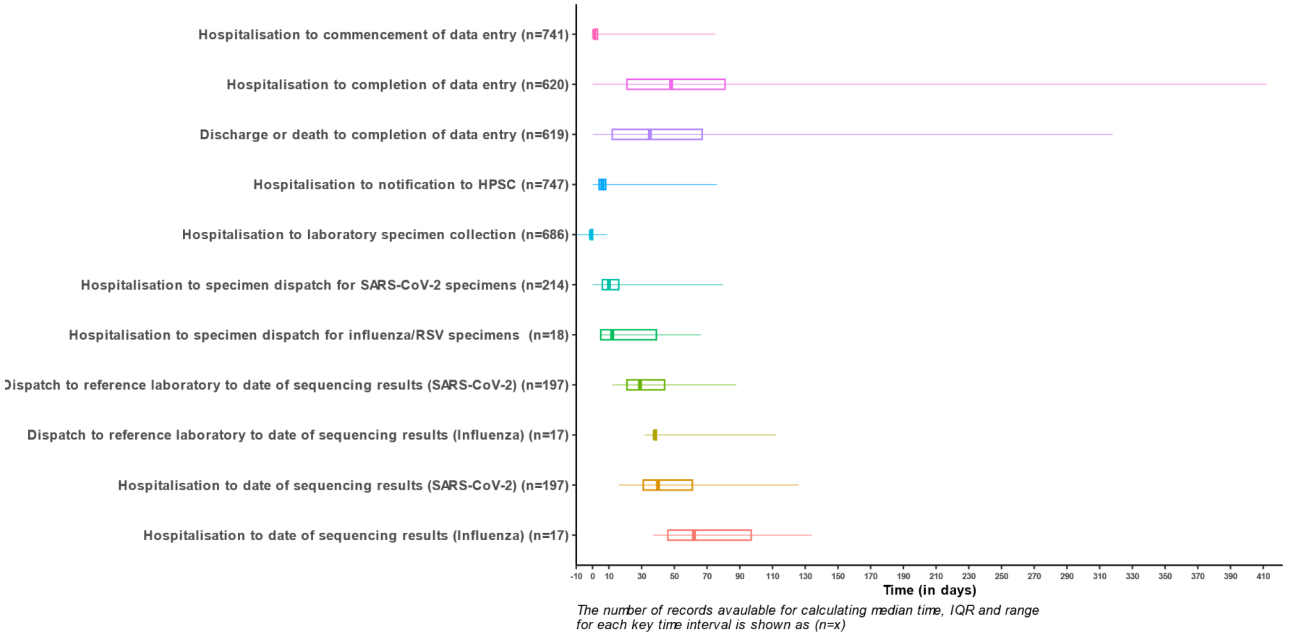


Fig. 2 Median time (IQR and range) for key intervals in the surveillance pathway

validation and reporting. The need for streamlining, protocols and automation was identified.

"There were lots of processes to put in place. SARI surveillance is person dependent, IT dependent and lab capacity dependent, but that is the complex surveillance system that it is." (KII1).
"The same chart may have to be requested multiple times to enter the data, you might not be able to have access to the chart and then you cannot report and then there is a delay." (FGD2, P1).
"It needs to be more streamlined; from an efficiency point of view" (FGD2, P2).
"Ideally SARI surveillance should be an automated surveillance system; without being electronic it is really manual and dependent on key people." (KII1).

Flexibility

Sub-theme: a reliable system that faces a challenge in surge periods

All participants reported that the SARI surveillance system is reliable. However, participants reported that surge periods have challenged the system, impacting on reliability, particularly the timeliness of data collection, collation and reporting.

"SARI data from this single site accurately tracks what we are experiencing on the ground" (KII5).
"The Omicron surge was challenging, if we had another variant of concern that was as transmissible but more severe, we may struggle to cope." (FGD1, P2).
"If the purpose of SARI surveillance is to identify new variants quickly, the current system is probably not robust enough to do that." (KII3).

Sub-theme: need for extensive resources and skilled staff

Due to the complex nature of the surveillance system extensive resources are required, specifically skilled, trained professionals.

"SARI surveillance is dependent on staff; staffing is the biggest challenge; in that way it is not robust." (KII3).

Acceptability

Sub-theme: importance

SARI surveillance is an important surveillance system that adds to the information reported by existing surveillance systems.

"During the COVID-19 pandemic, it became clear that there were gaps and in particular, a disconnect between a positive test and morbidity, SARI surveillance fills that gap, and it is therefore critical. SARI surveillance is essential for pandemic responsiveness and SARI data are important for healthcare planning." (KII5).

Sub-theme: acceptable in the clinical site and aligns with clinical priorities

SARI surveillance is acceptable within the participating hospital; adequate resources and staffing were identified as crucial to acceptability.

"SARI surveillance fits in well with existing pathways and is aligned with clinical requirements." (KII4).

Sub-theme: communication and relationships

Relationships between clinical and public health professionals were identified as being key to the establishment and sustainability of the surveillance system.

"Communication is crucial, especially if there was to be an emerging pathogen." (KII3).
"There were pre-existing relationships and so the expansion to SARI surveillance made sense, there were good affiliations and buy-in from the beginning." (KII5).

Usefulness

Among respondents to the web-based questionnaire, 79% (59/75) reported that they read the SARI surveillance report every week or most weeks for information, to monitor and anticipate SARI trends, to provide a general overview of SARI activity and communication and to facilitate engagement with relevant stakeholders to inform public health action/advice and health services planning. In free-text responses, some participants suggested that shortening the report or real-time reporting via a dashboard, would facilitate easier communication, engagement, and action.

Of respondents, 64% (37/58) strongly agreed/agreed that the surveillance system is meeting its objectives (Fig. 3). Among those who disagreed, their primary concern was lack of representativeness in a single site.

Among those who disagreed that that the surveillance system has the ability to detect unexpected events (7.3%, 4/55) (Fig. 3), concerns included the ability of the system to detect unusual events and the need for triggers to facilitate recognition of unusual events e.g. a high proportion of cases occurring in younger age groups. Timely sequencing was also identified as being essential

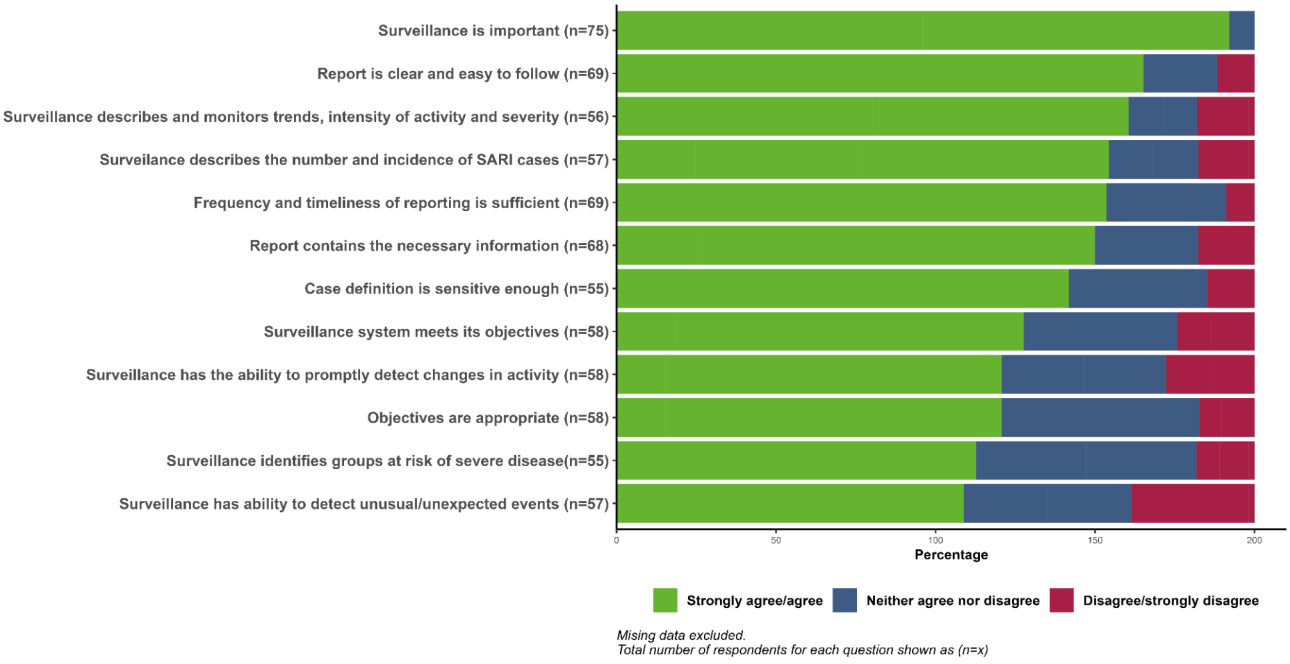


Fig. 3 Results of web-based questionnaire

Table 3 Proportion of notified SARI cases also meeting WHO SARI case definition

Symptoms	WHO SARI case	
	n	%
Fever (*n* = 747)	354	47
Cough (*n* = 746)	603	81
Onset within the last 10 days (*n* = 747)	691	93
Case meets WHO case definition	256	34

to facilitate the detection of unusual events. Overall, 96% (72/75) strongly agreed/agreed that SARI surveillance is important (Fig. 3).

Sensitivity of the case definition

All SARI cases included in the SARI database met the adapted ECDC SARI case definition (*n* = 747). Of these, 34% met the WHO SARI case definition (Table 3).

In the web-based questionnaire, 71% (39/55) agreed/strongly agreed that the adapted ECDC SARI case definition was sensitive enough to detect cases, monitor trends and to detect unusual events (Fig. 3). In the qualitative study, some participants reported that the case definition may be oversensitive, however, others reported that it is likely that some SARI cases may be missed. Some participants expressed the need for clarity about the case definition in terms of what exactly the SARI surveillance system is trying to capture.

> *"We have probably erred on the oversensitive." (KII2).*
> *"Under-ascertainment is probably happening; we may be missing atypical cases." (KII5).*
> *"I am worried about the case definition and what exactly we are trying to capture. What do we mean by SARI, it is all hospitalisations as a consequence of a respiratory pathogen?" (KII4).*

Discussion

The establishment of SARI surveillance in Ireland has strengthened national surveillance capacity and this evaluation has found that the current single-site SARI surveillance system is a complex but acceptable system that produces high-quality, complete and timely data that, in conjunction with other national surveillance systems, is useful to inform public health action, advice and communication and health service planning [19, 20].

The COVID-19 pandemic highlighted the importance of data to inform public health action and exposed global weaknesses in multiple aspects of public health intelligence, including surveillance capacity [21, 22]. A key objective of the National Health Protection Strategy for Ireland 2022–2027 is to strengthen surveillance and epidemiological analysis of health protection threats [23]. Public health decision-making at local, national, regional

and global levels must be based on real-time, accurate data and analysis [21, 24]. To achieve this, surveillance systems must be supported by adequate staffing, robust IT infrastructure, and timely access to diagnostics [21, 25–27]. This evaluation has identified key priorities to guide both improvements within the existing SARI surveillance system and the expansion.

To improve the simplicity, timeliness and acceptability of the surveillance system, this evaluation has identified a need for clear and robust operational protocols and streamlined and automated operational procedures. A priority for improving flexibility, timeliness and acceptability is the provision of adequate resources; specifically, skilled staff, to allow the system to function effectively and efficiently at all times. This evaluation has found that surge periods of infection have challenged the ability of the surveillance system to produce timely data. The development of a SARI surveillance surge plan, including the identification of additional resources and the development of a minimum core SARI dataset, would ensure the continued reliability of the surveillance system during surge periods and improve flexibility.

In this evaluation, information required for completion of certain variables e.g. ethnicity, was not systematically or reliably recorded in the medical notes, resulting in sub-optimal completeness. The collection of equity stratifiers, including ethnicity, should be reviewed and expanded within the SARI surveillance system to allow for a more comprehensive assessment of the burden of disease and an assessment of the impact of health inequality on SARI infection rates. This would be useful to inform public health action and targeted prevention. Sub-optimal completeness of vaccination status variables in this evaluation emphasises the need for data linkage and integration; use of an individual patient identifier is a key enabler of this.

The International Health Regulations (2005) require national surveillance systems to be capable of timely detection, assessment and analysis of epidemiological data, including laboratory results for informed decision-making [25, 26]. Robust and accurate surveillance data, including genomic data, are crucial to all aspects of health threats preparedness and planning, particularly in the identification of novel pathogens [28–30]. In this evaluation, SARI cases were recognised, and specimens were collected in a timely manner, however, the transfer of specimens and receipt of sequencing results from the reference laboratory during the evaluation period was not sufficiently timely to support the collation and analysis of surveillance data. Since completion of this evaluation, on-site SARS-CoV-2 WGS is now conducted in the participating hospital which has improved timeliness; on-site WGS should be a priority for SARI surveillance sites to improve timeliness of results.

The analysis of the sensitivity of the SARI case definition undertaken in this evaluation has indicated that the WHO case definition was less sensitive when compared with the adapted ECDC SARI case definition. Only one third of notified SARI cases met the WHO case definition. This finding aligns with concerns expressed at the clinical site at the commencement of SARI surveillance that the WHO case definition was not sensitive enough to detect all patients presenting with SARI [4]. However, opinions on the sensitivity of the case definition among participants in this evaluation varied. Some participants expressed concern that the case definition may be too sensitive while others felt that under-ascertainment of cases may be occurring. Ongoing review and clinical audit of the SARI case definition in conjunction with European partners along with training and clear clinical guidelines for participating SARI hospital sites regarding the application of the case definition is essential to ensure accurate and consistent SARI case recognition nationally and internationally.

This evaluation identified that collaboration and communication have been essential for the acceptability of the SARI surveillance system. A key enabler to ensuring improvements within the existing system and to ensuring a successful expansion is buy-in and engagement from all key stakeholders in participating sites. A national SARI steering committee with clinical and public health leadership providing oversight and governance should be established to guide expansion. Lack of representativeness and generalisability of the current single-site SARI surveillance system was identified as a concern in this evaluation, limiting acceptability among stakeholders. The expansion of SARI surveillance to create a SARI sentinel hospital network aims to address this. Expansion should therefore improve the acceptability and usefulness of the SARI surveillance system.

Nationally and internationally, as COVID-19 testing policies continue to adapt with less community testing and with multi-pathogenic winters (with co-circulation of SARS-CoV-2, influenza and respiratory syncytial virus), SARI surveillance and other sentinel surveillance systems e.g. for primary care, have increased in importance and are essential components of surveillance in order to detect changing trends and threats. Internationally, sentinel surveillance systems will require prioritisation, funding, and resource allocation to function as early warning systems. This evaluation identified priorities for expansion which are being implemented in Ireland to inform the creation of a SARI sentinel hospital network. These findings can be used by other countries in the wider European region and internationally to guide expansion of existing SARI surveillance systems and may also be helpful to inform countries who are establishing SARI surveillance.

Limitations

This evaluation did not evaluate all attributes of the SARI surveillance system. Specifically, cost and representativeness were not evaluated. While the findings of the evaluation will be helpful to inform the expansion of SARI surveillance nationally, the experience in one SARI site may not be generalisable to all potential future hospital sites.

The response rate to the web-based questionnaire was low (21%) and even among those who responded, the non-response to individual questions was high. While there was representation from all the main professional groups, the low response rate limits the representativeness and generalisability of the evaluation findings.

Conclusion

The SARI surveillance system has been an important component of national surveillance since its establishment in 2021 and has contributed to strengthening national surveillance capacity in Ireland. This evaluation has found that the current single-site SARI surveillance system produces accurate, high-quality, and timely data which are used to guide public health action. Improvement in the timeliness of WGS data, recording of vaccination status and equity stratifiers e.g. ethnicity, would strengthen data quality. The planned expansion to create a SARI sentinel hospital network should be a priority for the

Health Service in Ireland. The primary considerations for SARI expansion include process automation, standardised operating protocols across participating sites, adequate funding and resource allocation and strong clinical and public health leadership and governance. This will improve the flexibility, acceptability, usefulness and representativeness of the system and allow SARI surveillance to function reliably at all times, including during epidemic or pandemic surge periods.

Abbreviations

CDC	Centers for disease control and prevention
COREQ	Consolidated criteria for reporting qualitative research
ECDC	European centre for disease prevention and control
HPSC	Health protection surveillance centre
HSE	Health service executive
IQR	Interquartile range
SARI	Severe acute respiratory infection
WGS	Whole genome sequencing
WHO	World health organization

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12889-025-21645-3.

Supplementary Material 1

Supplementary Material 2

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Author contributions

LM, LD, RD and JOD contributed to the design of the evaluation. RD contributed to the acquisition of the data for the evaluation. LM analysed and interpreted the data and drafted and revised the manuscript. LD, RD, JOD contributed to drafting and revising the manuscript. All authors read and approved the final manuscript.

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Data availability

The datasets analysed during the study are from the national Severe Acute Respiratory Infection (SARI) surveillance database in Ireland and are not publicly available to protect confidentiality. Weekly published SARI reports are available at www.hpsc.ie.

Declarations

Ethics approval and consent to participate

This evaluation was part of routine operational practices conducted in the interest of public health. Ethical approval or patient consent was not required for any component of this evaluation as under medical officer of health legislation in Ireland and the Health Act 2007, service evaluations and statutory public health work conducted for the planning and delivery of health services do not require research ethical approval [31, 32]. The evaluation was undertaken in accordance with the Health Service Executive (HSE) data protection policy and Article 9 of the General Data Protection Regulation (GDPR) 2018, which provides exception to the prohibition of the processing of sensitive personal health information without consent for reasons of public interest in the area of public health [33].

Consent for publication

All authors consent to publication of this work.

Competing interests

The authors declare no competing interests.

Disclaimer

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